nature genetics

Article

https://doi.org/10.1038/s41588-026-02573-3

Mosaic integration of spatial multi-omics with SpaMosaic

Received: 22 November 2024

Accepted: 11 March 2026

Published online: 24 April 2026

Check for updates

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With the advent of spatial multi-omics, mosaic integration of diverse datasets with partially overlapping modalities enables the construction of comprehensive multimodal spatial atlases from heterogeneous sources. Here we present SpaMosaic, a tool that uses contrastive learning and graph neural networks to build a modality-agnostic, batch-corrected latent space for spatial domain identification and missing-modality imputation. We systematically benchmarked SpaMosaic against existing integration methods using simulated data and experimentally acquired datasets spanning RNA and protein abundance, chromatin accessibility and histone modifications from brain, embryo, tonsil and lymph node tissues. SpaMosaic consistently outperformed other methods in identifying coherent spatial domains by reducing noise and mitigating batch effects. We further challenged SpaMosaic with heterogeneous real-world datasets spanning different technologies, developmental stages, resolutions and modality compositions, where it consistently resolved fine anatomical structures and enabled comprehensive mouse embryo atlasing. Beyond integration, SpaMosaic enables accurate imputation of missing modalities. In a mosaic mouse brain dataset, the imputed histone modifications not only recapitulated expected transcriptome–epigenome correlations but also uncovered more region-specific regulatory links compared to the measured chromatin accessibility data, demonstrating the ability to infer relationships across modalities without coprofiling. Computationally, SpaMosaic is highly scalable, capable of integrating over 100 sections and processing a single section with more than 800,000 spots. In summary, SpaMosaic provides a versatile framework for unifying the rapidly accumulating heterogeneous spatial omics data into comprehensive biological atlases.

Spatial omics are powerful techniques for studying cellular heterogeneity in its spatial context within tissues[1]. Current developments in spatial multi-omics are expanding the scope to simultaneously acquire multiple omics from one tissue section[2]. For example, spatial ATAC–RNA-seq captures chromatin accessibility and RNA expression[2], spatial ATAC–RNA–Protein-seq enables tri-omics profiling[3] and spatial Mux-seq[4] can simultaneously profile up to five different modalities. By capturing different omics profiles from the same sample, these complementary views can be used to construct more comprehensive pictures and to better dissect phenotypic niches[5]. However, spatial omics experiments are technically challenging, involving trade-offs among resolution, sensitivity and throughput, and acquiring multiple omics from the same section exacerbates these challenges. Currently, the most commonly available data are from single-omics modality experiments on different, sometimes adjacent, sections. Thus, spatial multi-omics with shared but also complementary omes can be

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computationally integrated to produce a more coherent, complete and higher-dimensional picture of tissue biology. Such integration, termed mosaic integration, can help consensus domain identification, enhance signal-to-noise ratio, impute missing assays in different sections and dissect cross-linking between omics, which will be crucial for constructing multi-omics spatial tissue atlases.

Essentially, mosaic integration seeks to integrate multiple batches with different multimodal compositions[6]. However, unlike diagonal integration with no shared batches or modalities, mosaic integration leverages a mosaic of shared modalities across batches to serve as bridges for data integration. Therefore, mosaic integration accomplishes both vertical integration by harmonizing modality heterogeneity within the same cells and horizontal integration by reducing technical variations within the same modality across batches[7]. In the context of mosaic integrating spatial omics data, the spatial context is an additional information layer that should be used to accomplish a more coherent and accurate integration.

There are currently several single-cell mosaic integration methods available, like Cobolt[8], scMoMaT[7], StabMap[9] and MIDAS[10]. As they are tailored for single-cell mosaic integration, they do not consider the spatial context and may struggle with spatial data. For spatially aware methods, there are neighborhood-aware clustering approaches like BANKSY[11], horizontal integration methods like SEDR[12], GraphST[13], STAligner[14] and STitch3D[15], and vertical integration tools like SpatialGlue[5]. Due to their respective assumptions on data availability, none of them is usable for spatial mosaic integration. While CellCharter[16] and PRESENT[17] are designed for vertical and horizontal spatial integration, they cannot handle the general case of spatial mosaic integration.

Here we present SpaMosaic for spatial mosaic integration. Combining contrastive learning[18] with graph neural networks (GNNs), SpaMosaic projects each section into a modality-agnostic latent space in a spatially aware manner while minimizing batch variations. Specifically, we adopted a weighted light graph convolution network (WLGCN) to encode the similarity across spots. The output latent embedding is usable for imputing data modalities and for downstream analyses, such as spatial domain identification and joint analysis of multiple sections. We first used simulated datasets to showcase SpaMosaic at mosaic integration while reducing technical noise and recovering hidden spatial patterns. We then benchmarked SpaMosaic with experimentally acquired data of human and mouse tissues with RNA, ATAC, histone modifications and protein modalities. Overall, SpaMosaic showed superior performance at modality alignment, batch integration and spatial domain identification. Furthermore, we demonstrated SpaMosaic's versatility in handling different configurations of modalities, resolutions, section sizes and technologies across diverse biological contexts, thereby enabling comprehensive and coherent analyses of heterogeneous spatial omics data.

Results

The SpaMosaic model

SpaMosaic integrates mosaic spatial omics data of multiple tissue sections to output a joint latent space for downstream analyses (Fig. 1a). For input, SpaMosaic takes in mosaic feature-by-spot (or cell) count matrices and associated spatial coordinates (Fig. 1b). SpaMosaic is acquisition technology agnostic and can integrate sections of different sizes and spatial resolutions. To capture the links between modalities, the modalities must be directly or indirectly connected through one or more sections—a direct connection appears in the same section, while an indirect connection requires multiple intermediary direct connections. Within each section and modality, SpaMosaic first performs dimension reduction on highly variable features and removes batch effects using horizontal integration methods such as Harmony[19] and Seurat[20]. SpaMosaic then constructs a heterogeneous adjacency graph for each modality to capture both spatial distance-based adjacency (nearest spatial neighbors within a section) and omics feature-based adjacency (mutual nearest neighbors (MNNs)[21] across sections). To improve robustness, we leverage the spatial context and incorporate outlier-detection algorithms to filter out noisy MNN pairs.

Next, SpaMosaic builds a modality-agnostic latent space using WLGCNs in a contrastive learning framework (Supplementary Fig. 1a). For each modality, a WLGCN encodes modality-specific information from the modality inputs and adjacency graph into embeddings. We use contrastive learning to align modality-specific embeddings—embeddings from different modalities at each spot form positive pairs, while embeddings from different spots form negative pairs, irrespective of modality. This seeks to harmonize different modalities while preserving variation across spots. After alignment, the modality-specific embeddings are averaged to produce the final multimodal embedding. For single-modality sections, we use a feature reconstruction objective to preserve interspot variations. SpaMosaic outputs integrated spot embeddings across sections, with imputed feature matrices for missing omics data, enabling downstream analyses such as spatial domain identification and cross-modality feature correlation analysis. To impute, SpaMosaic uses feature information from nearest neighbors in the modality-aligned latent space (for details, see Methods and Supplementary Fig. 1b).

Benchmarking on simulated data and multitissue experimental datasets

For benchmarking, we simulated five sets of spatial multi-omics datasets (three replicates each) following ref. 22, configured in a mosaic arrangement comprising one paired RNA–protein section, one RNA-only section and one protein-only section (Extended Data Fig. 1a,b; Methods). Each simulated section has a ground truth of four spatial factors and a background (Fig. 2a). These simulations incorporated batch effects and varying noise levels to mimic experimental data (Extended Data Fig. 1c). Notably, we specifically depleted RNA expression in factor 1 and protein expression in factor 4 to test the ability to reconstruct comprehensive spatial factors by integrating disjointed modalities. We evaluated SpaMosaic and its ablation variant, SpaMosaic (nonspatial; Methods), against five state-of-the-art mosaic integration methods (Cobolt, scMoMaT, StabMap, CLUE, MIDAS) and two spatial domain identification methods (BANKSY, CellCharter) that served as single-modality spatial clustering baselines. We evaluated four performance aspects—spatial factor preservation using adjusted Rand index (ARI), identified spatial domain's continuity using percentage of abnormal spots (PAS)[23] and spatial chaos (CHAOS) score[23] metrics, batch mixing using integration local inverse Simpson's index (iLISI)[24] and modality alignment using the F1 score computed for label transfer accuracy[10]. Higher scores indicate better performance for all metrics, except for PAS and CHAOS, where lower scores denote better performance.

We first examined single-modality baselines using one replicate from dataset 1. Leiden clustering (with Harmony for batch integration) captured all four factors only when applied to the concatenated data of section 1 (Fig. 2b). In the RNA modality (section 2 and section 1-RNA), Leiden identified factors 2, 3 and 4 but failed to separate the depleted factor 1 from the background. In the protein modality (section 3 and section 1-protein), it also captured factors 1, 2 and 3 but not the depleted factor 4, as confirmed by Uniform Manifold Approximation and Projection (UMAP) plots where these depleted factors merged with the background (Fig. 2c). For spatial domain methods, we applied BANKSY to the RNA modality, and CellCharter to both RNA and protein. In the RNA modality, both methods captured factors 2, 3 and 4; notably, while CellCharter yielded more coherent domains, it incorrectly split factor 2 into two clusters (Extended Data Fig. 1d). In the protein modality, CellCharter failed to clearly delineate factor 4. UMAP visualization further supported these observations, revealing overlaps between the background and spatial factors (Extended Data Fig. 1e).

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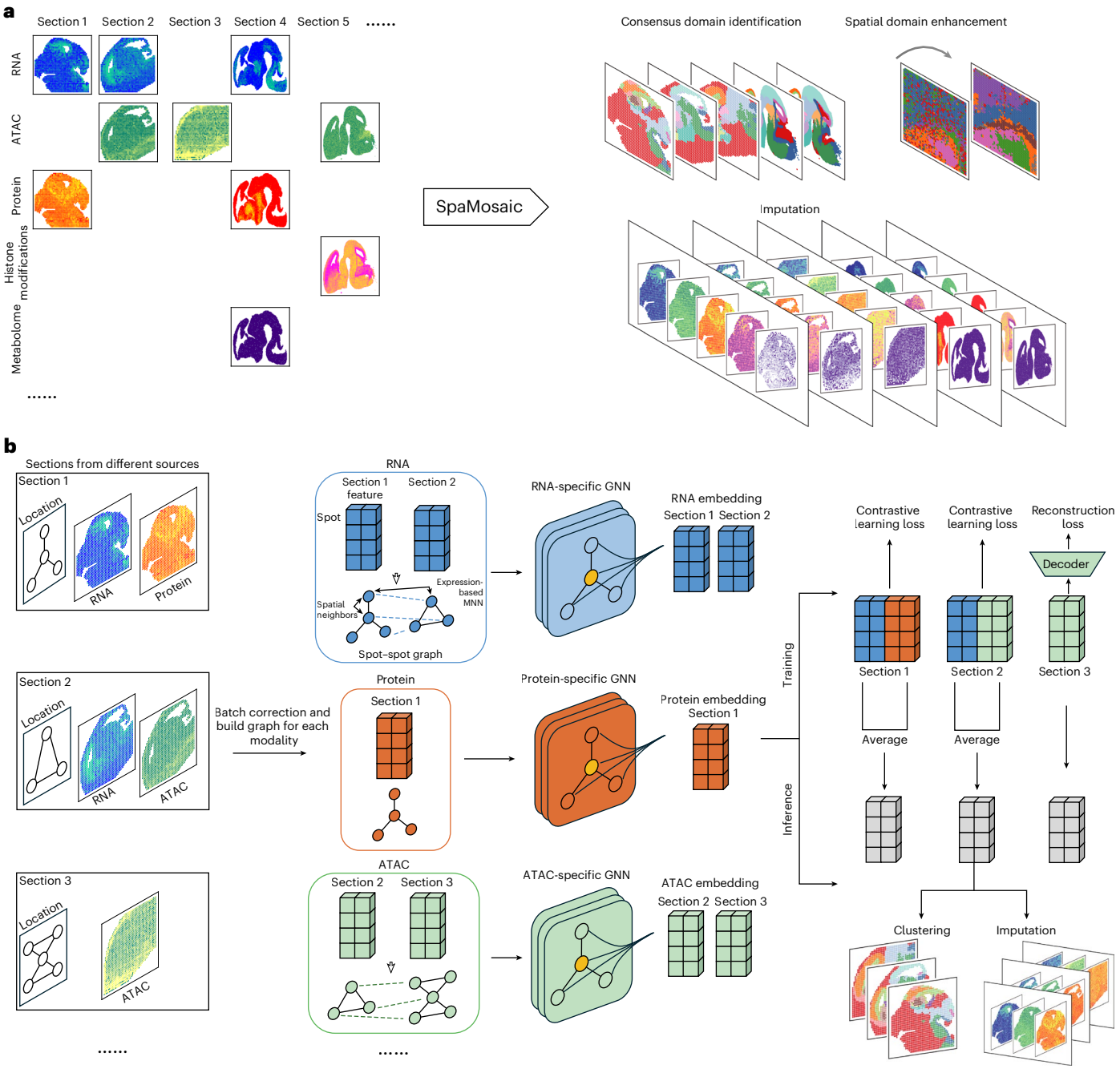


Fig. 1 | Overview of SpaMosaic framework. a, SpaMosaic integrates multiple tissue sections with diverse modality compositions into a unified representation. It enables the identification of consensus spatial domains across sections, imputation of missing modalities and enhancement of spatial domain resolution by revealing coherent anatomical structures that are obscured in single-modality analyses. Left, an abstract representation of the general mosaic integration task, where sections with different modality compositions can be integrated. **b**, Illustration of the SpaMosaic workflow using a specific example of spatial multi-omics mouse brain dataset integration. SpaMosaic first reduces data dimensionality and then performs batch effect correction to generate batch-corrected low-dimensional representations for each modality. Thereafter, a heterogeneous adjacency graph is constructed across spots within each modality. SpaMosaic uses GNNs to embed the modality-specific representations into modality-specific embeddings. The paired multimodal embeddings serve as input to contrastive learning, which aims to align embeddings from different modalities and construct a modality-agnostic latent space. Finally, SpaMosaic infers the aligned embeddings for each section for downstream analysis (such as spatial domain identification) and imputation.

While it was unsurprising that factor 1 (RNA) and factor 4 (protein) were not recovered, their identification of other factors suffered from the noise present.

We then considered the results from mosaic integration methods. SpaMosaic recovered all four spatial factors across all three sections with reduced noise and superior shape accuracy compared to single-modality baselines (Fig. 2d). In contrast, while most competing methods delineated all factors in section 1 (except MIDAS, which failed on factor 3; Extended Data Fig. 2a), their performance dropped in the unimodal sections. Specifically, in sections 2 and 3, factors 2 and 3 were consistently recovered with substantial noise. Notably, for the missing factors, factor 1 in section 2 and factor 4 in section 3, only CLUE achieved effective recovery, whereas Cobolt and scMoMaT showed partial success for factor 4. UMAP embeddings validated these results—SpaMosaic demonstrated precise integration of all three sections within each factor's cluster while maintaining clear spatial separation consistent

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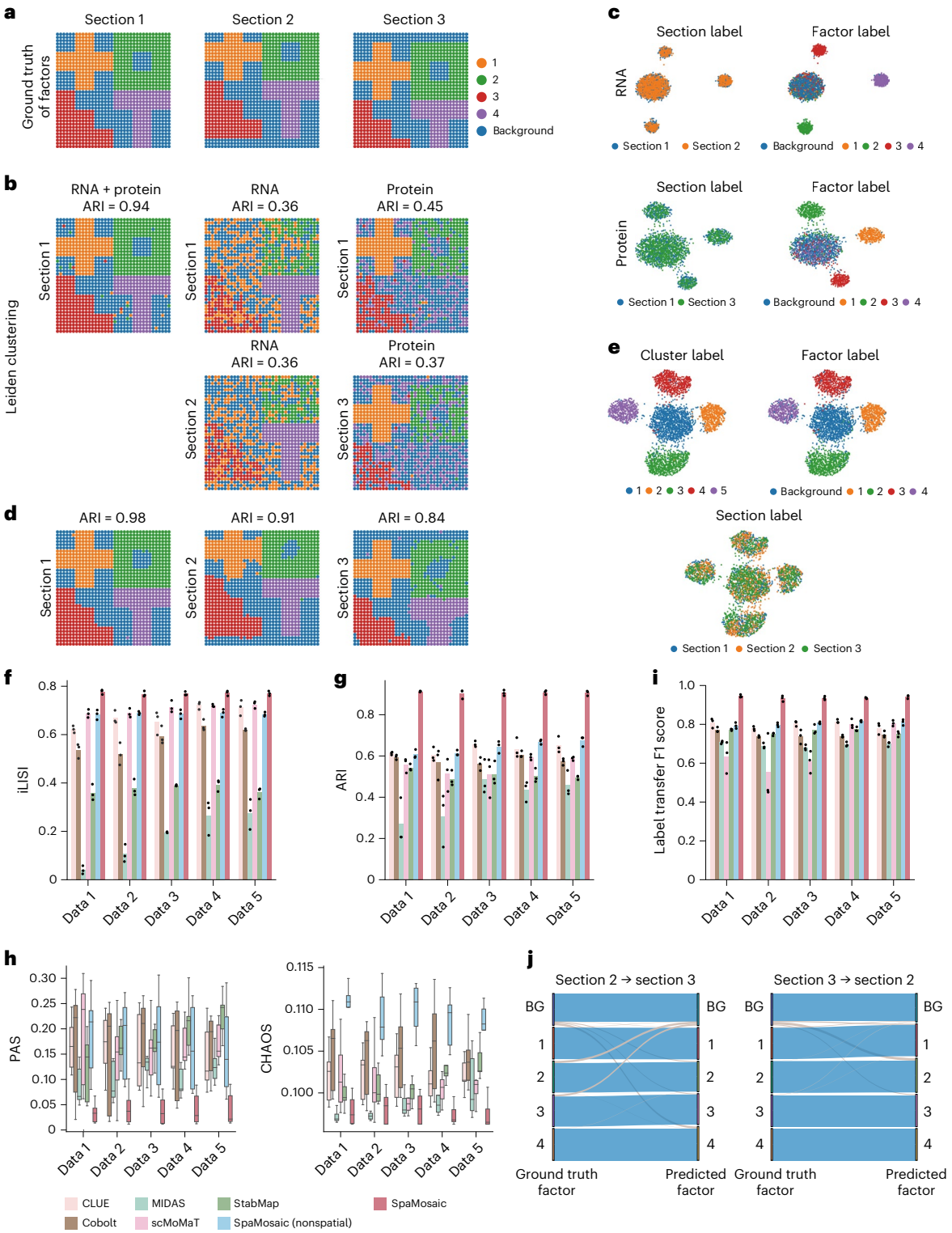


Fig. 2 | Benchmarking on simulated datasets. a, Ground truth of spatial factors and background of the three sections. **b**, Clustering patterns obtained with the concatenated RNA and protein profiles, RNA profiles and protein profiles, respectively. RNA features from sections 1 and 2 were jointly clustered using the Leiden algorithm, as were protein features from sections 1 and 3. Multimodal clustering of section 1 was performed on concatenated RNA and protein features using the Leiden algorithm. ARI scores were evaluated by comparing the clustering with the ground truth. **c**, UMAP plots of batch-corrected RNA (sections 1 and 2) and protein (sections 1 and 3) profiles. Spots are colored by section (left) or factor (right). **d**, Clustering patterns obtained using SpaMosaic's embeddings for all three sections. **e**, UMAP plots of SpaMosaic's embeddings. Spots are colored by clustering, factor or section. **f**, iLISI scores across five simulated datasets. Bars represent the mean values, with overlaid dots indicating individual replicates ($n = 3$). **g**, ARI scores across five simulated datasets. Bars represent the mean values, with overlaid dots indicating individual replicates ($n = 3$). **h**, PAS and CHAOS scores of benchmarked methods across five simulated datasets. Box plots display the distribution of scores derived from $n = 3$ independent replicates, each consisting of scores on three slices (total $n = 9$ data points per box). Centerlines indicate the median; box limits represent the IQR; whiskers denote 1.5× IQR. **i**, Label transfer F1 scores across five simulated datasets. Bars represent the mean values, with overlaid dots indicating individual replicates ($n = 3$). Each replicate data point is calculated as the average of reciprocal transfer scores between two single-modal slices. **j**, Sankey plots showing the comparison of prediction and ground truth of factors on the first replicate of the first simulation setting. BG, background; IQR, interquartile range.

with ground truth (Fig. 2e), whereas competing methods achieved only varying degrees of integration (Extended Data Fig. 2b).

Using quantitative metrics, we compared SpaMosaic and single-cell mosaic integration methods on all simulated datasets. SpaMosaic consistently outperformed across all datasets with the highest scores in iLISI (batch integration), ARI (preservation of spatial factors) and PAS (continuity of identified spatial domains), followed by scMoMaT and CLUE (Fig. 2f–h). While MIDAS marginally led in the CHAOS metric on datasets 1–3, SpaMosaic ranked first or second overall and dominated section-specific evaluations, specifically sections 1 and 2 (Extended Data Fig. 2c,d). Taken together, these results quantified SpaMosaic's capture of spatially coherent and accurate factors, which we attribute to its incorporation of spatial information, batch effect removal capabilities and integration of information from all sections to recover the missing factors. Next, cross-modal alignment was assessed through label transfer. Here the labels from one single-modality section were used to predict the corresponding labels in another single-modality section, testing whether each method could effectively align and transfer information across modalities[10]. SpaMosaic achieved the highest median F1 scores across all datasets (Fig. 2i), followed by the nonspatial variant and CLUE. Using Sankey plots (first replicate of dataset 1), we found that SpaMosaic's misassignments were primarily restricted to factor-background ambiguity rather than confusion between distinct spatial factors (Fig. 2j), consistent with the clustering results (Fig. 2d,e).

We then performed evaluation using experimentally acquired datasets, with selective modality masking to simulate mosaic integration scenarios. We assessed four datasets spanning different tissue types, developmental stages and modality combinations—(1) embryonic mouse brain sections (E13.5–E18.5) with RNA and ATAC (Misar-seq[25]), (2) postnatal mouse brain with RNA and H3K4me3 (refs. 1,2), (3) human lymph node with RNA and protein, and (4) human tonsil with RNA and protein. Overall, SpaMosaic successfully resolved fine anatomical structures and maintained spatial continuity across all scenarios, outperforming competing methods that often merged distinct regions or produced inconsistent cluster assignments (Supplementary Note 1).

Reconstruction of comprehensive multi-omics profiles through mosaic integration and imputation

Beyond spatial domain identification, SpaMosaic can impute unmeasured modalities to reconstruct comprehensive multi-omics profiles for downstream analyses. Benchmarking against MIDAS, BABEL[26], totalVI[27] and MultiVI showed that SpaMosaic consistently excelled in preserving both spot–spot and feature–feature correlations (Supplementary Note 2).

Here we applied SpaMosaic to a complex postnatal mouse brain dataset comprising four sections acquired through two sequencing technologies (spatial RNA–ATAC–seq and spatial RNA–histone modification cosequencing spRA + spRH). These sections share RNA as a bridging modality (with batch effects; Supplementary Fig. 2a) but profile diverse epigenomic measurements (H3K27me3, H3K4me3, H3K27ac, ATAC; Fig. 3a). This technical heterogeneity across sections challenges integration yet uniquely enables correlation inference among distinct epigenetic layers that were not physically co-assayed.

We first verified that SpaMosaic accurately aligned anatomical structures across these sections, establishing a foundation for downstream imputation. Specifically, SpaMosaic's clustering successfully recapitulated anatomical structures matching the Allen Brain Atlas[28], accurately resolving regions like the diagonal band nucleus (NDB), corpus callosum genu (CCG) and lateral septum (Fig. 3b and Supplementary Fig. 2b). These identities were validated by region-specific expression of known markers, including *Mbp* (CCG), *Stard5* (ACB) and *Fezf2*/*Cux2* (deep/upper cortical layers; Fig. 3c). Notably, SpaMosaic clearly distinguished the ACB from the caudoputamen (CP) and stratified cortical layers, structures consistently unresolved by competing methods (Cobolt, scMoMaT, StabMap, UINMF). Quantitatively, SpaMosaic excelled in spatial continuity (PAS/CHAOS; Fig. 3d). While Cobolt yielded higher iLISI scores (Fig. 3e), this stemmed from improper merging of distinct regions. Thus, SpaMosaic achieved an optimal balance between batch integration and anatomical preservation (for detailed comparisons, see Supplementary Note 3).

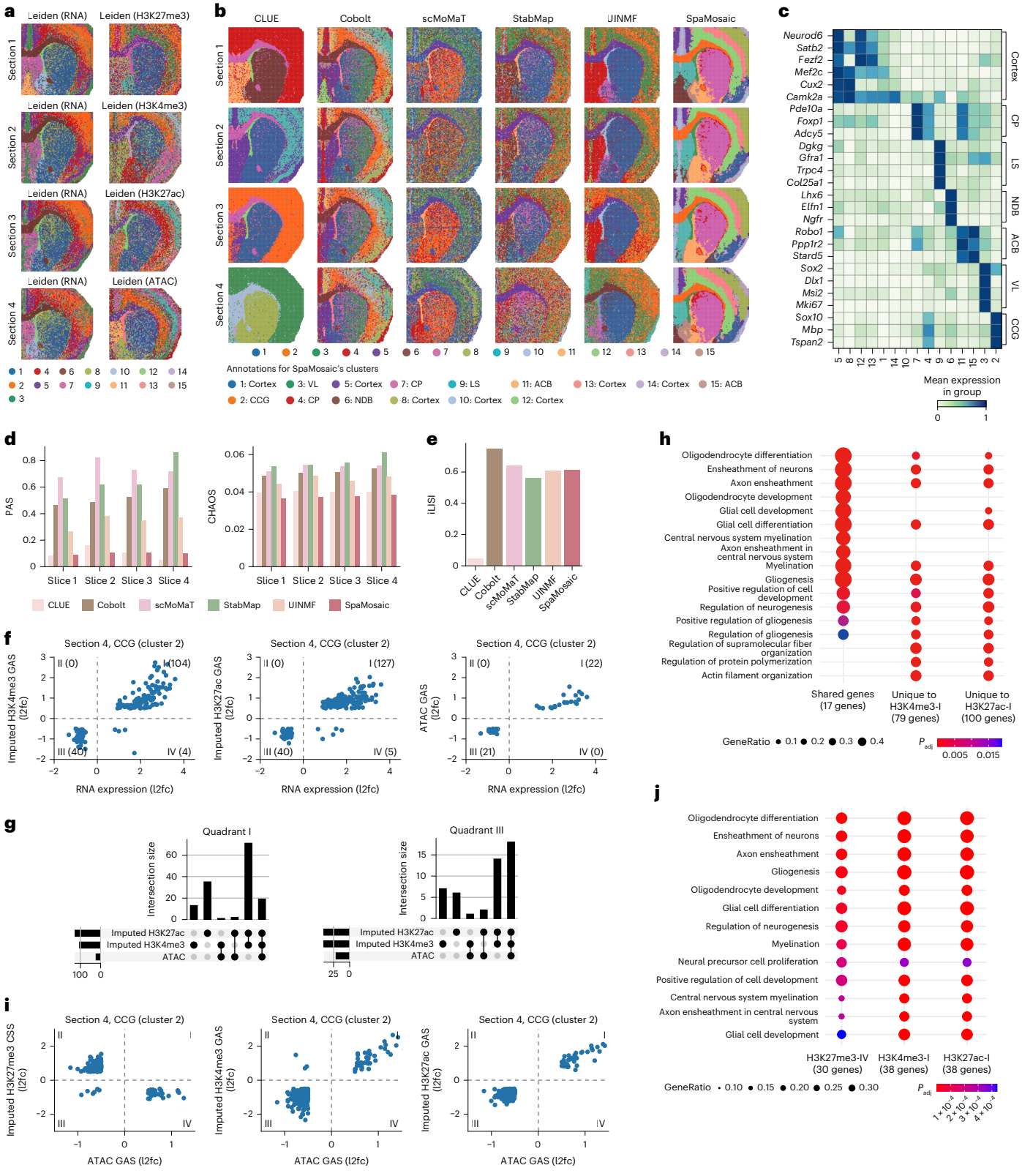
We then imputed the missing epigenomic profiles and investigated their cross-modal correlations by comparing gene activity scores (GAS) or chromatin silencing scores (CSS)[29] of measured versus imputed profiles in a cluster-specific manner. We retained only region-specific genes with an adjusted P value $<10^{-5}$ and absolute $\log_2$ fold change (FC) >0.5 (Extended Data Fig. 3a–d). For the CCG cluster, the measured H3K27me3 CSS showed expected anticorrelation with RNA (as H3K27me3 is a suppressor), and the imputed H3K27me3 CSS showed similar anticorrelations across other sections (Extended Data Fig. 3a). The measured H3K4me3 GAS on section 2 positively correlated with RNA (matching H3K4me3's promoter function), and the imputed H3K4me3 GAS of other sections likewise showed positive correlations. Similarly, the imputed H3K27ac and ATAC modalities showed positive correlations with RNA. These patterns were consistent across other clusters, including VL (cluster 3), CP (cluster 7) and cortex (cluster 8; Extended Data Fig. 3b–d), demonstrating that SpaMosaic's latent representation captured and preserved cross-modal relationships during imputation.

To further investigate these imputations, we compared the correlations between measured RNA and GAS computed from measured ATAC versus imputed histone modifications on section 4. Using the same filtering ($P_{adj} < 10^{-5}$, $\log_2(FC) > 0.5$), the imputed histone modifications (H3K4me3 and H3K27ac) showed more genes correlated with RNA than measured ATAC GAS (Fig. 3f). Correlated genes (quadrants I and III) in the measured ATAC assay were also found in the imputed histone modification modalities (Fig. 3g), confirming that the imputed

Fig. 3 | Application of SpaMosaic to postnatal mouse brain dataset (spRA + spRH). a, Mosaic dataset configuration and clustering patterns of the four sections. The RNA cluster labels were obtained by performing Leiden clustering on the batch-corrected low-dimensional embeddings, and each epigenomic modality's cluster labels were obtained by performing Leiden clustering on the principal components of the epigenomics profiles. **b**, Spatial plots of clusters obtained from the methods for all four sections. Manual annotation was performed to identify SpaMosaic's clusters. **c**, RNA expression of brain region markers in SpaMosaic's clusters. **d**, PAS and CHAOS scores of benchmarked methods across the four sections. Bars represent metric values calculated independently for each tissue section ($n = 1$). **e**, iLISI scores of benchmarked methods. Bars represent the global score calculated across the integrated dataset ($n = 1$). **f**, Relationships among the imputed H3K4me3 GAS, imputed H3K27ac GAS and measured ATAC GAS with the RNA in the CCG region of the fourth section. Only region-specific genes were kept for all modalities if their adjusted P value was $<1 \times 10^{-5}$ and the absolute value of $\log_2$(FC) was >0.5. **g**, Upset plot of genes in quadrant I and quadrant III of all comparisons in **f**. **h**, Enriched GO terms for genes shared across quadrant I of all comparisons, as well as for genes unique to quadrant I of the H3K27ac and H3K4me3 comparison in **f**. Enrichment analysis was performed using a hypergeometric test, and P values were adjusted for multiple comparisons using the BH method. **i**, Relationships among the imputed H3K27me3 GAS, imputed H3K4me3 GAS and imputed H3K27ac GAS with measured ATAC GAS in the CCG region of the fourth section. Only region-specific genes were kept if their adjusted P value was $<1 \times 10^{-5}$ and the absolute value of $\log_2$(FC) was >0.5. **j**, Enriched GO terms for the genes in quadrant IV of H3K27me3 versus ATAC, quadrant I of H3K4me3 versus ATAC and quadrant I of H3K27ac versus ATAC comparisons in **i**. Enrichment analysis was performed using a hypergeometric test, and P values were adjusted for multiple comparisons using the BH method. BH, Benjamini–Hochberg.

https://doi.org/10.1038/s41588-026-02573-3

GAS preserved their relationships with measured gene signals. The overall stronger correlation between imputed histone modifications and RNA, compared to the ATAC–RNA correlation, indicates that, during model learning, RNA serves as the bridge modality explicitly aligned with all other modalities. We then examined the genes unique to the imputed histone modifications. Gene Ontology (GO) enrichment analysis of quadrant I genes unique to imputed H3K4me3 or H3K27ac revealed associations with oligodendrocyte differentiation and myelination (Fig. 3h), matching previous findings[2]. Across the three epigenomic modalities shown in Fig. 3f, 17 genes in quadrant I were shared, and they are highly expressed by oligodendrocytes that are enriched in the mouse brain's corpus callosum, including genes encoding key transcriptional regulators of oligodendrocyte development and differentiation (*Sox2* (ref. 30), *Sox10* (ref. 31), *Olig1* (ref. 32),



Zeb2 (ref. 33)) and genes encoding major myelin structural proteins (*Mobp*[32], *Plp1* (ref. 32), *Mag*[32], *Mal*[32]), which are promoted by H3K4me3 for expression[34,35]. Additional oligodendrocyte-enriched genes unique to H3K27ac or H3K4me3 quadrant I include genes encoding myelin structural proteins (*Ermn*[36], *Cnp*[32], *Mog*[32], *Opalin*[32]) and metabolic enzymes (*Enpp6* (ref. 37), *Sirt2* (ref. 38)). Quadrant IV genes showing inverse relationships between RNA expression and imputed histone mark's GAS included four genes for H3K4me3 (*Kif5a*, *Mt3*, *Mir682*, *Clu*) and five genes for H3K27ac (*Camk2n1*, *Igfbp5*, *Rps23*, *Rpl26*, *Lrrc17*), which have not been reported to be controlled by these histone marks and thus cannot not be verified. Consistent with these correlations, spatial mapping of the imputed histone modification GAS for CCG marker genes clearly demarcated the CCG with greater spatial coherence than the measured ATAC GAS (Supplementary Fig. 3), attributable to the information aggregation across sections and incorporation of spatial information during imputation. Similar signal enhancement was observed in other regions (VL, CP, cortex; Extended Data Fig. 3e).

We also investigated the relationships among different epigenomic modalities not comeasured in the same section. Analyzing section 4's CCG cluster, we examined the correlations between imputed histone modification assays and measured ATAC GAS. With the same filtering ($P_{adj} < 10^{-5}$, $\log_2(FC) > 0.5$), the imputed histone mark GAS or CSS showed expected correlations with the measured ATAC GAS—anticorrelation with repressor H3K27me3, and correlation with promoters H3K27ac and H3K4me3 (Fig. 3i). GO enrichment of anticorrelated genes (quadrant IV, H3K27me3) and correlated genes (quadrant I, H3K4me3 and H3K27ac) revealed shared processes including oligodendrocyte differentiation, development and myelination (Fig. 3j). The 29 genes shared among quadrant IV of H3K27me3 and quadrant I of both H3K4me3 and H3K27ac included genes encoding oligodendrocyte transcription factors (*Sox10*, *Sox2*, *Olig1*, *Olig2* (ref. 32)) and myelin sheath constituents (*Cldn11* (ref. 32), *Fa2h*[39], *Ugt8a*[40]). Notably, *Mbp* and *Plp1*, which encode major myelin structural proteins, were absent from this set, suggesting that their regulation may involve other mechanisms, such as DNA methylation[41]. Likewise, CP (cluster 7) and cortex (cluster 8) showed similar correlation patterns with overlapping GO terms (Supplementary Fig. 4), validating SpaMosaic's accuracy.

Overcoming cross-platform disparities for coherent real-world integration with SpaMosaic

We then challenged SpaMosaic with a scenario representative of complex real-world integration analyses—mosaic data of embryonic mouse brain acquired at six developmental stages. The dataset comprised three embryonic mouse brain sections (E13.5, E15.5, E18.5) profiled by Misar-seq (RNA and ATAC)[25], and three additional samples (E12.5, E14.5 and E16.5) with RNA profiles obtained using Stereo-seq[42] (Fig. 4a). These technologies differ markedly in resolution and data density. Stereo-seq has nanoscale capture (0.5-µm center-to-center spot spacing) that, at binning of 50, gives 15,000 bins and a 25-µm resolution, while Misar-seq has a ~2,000 grids capture and 50-µm grid resolution (Fig. 4b,c). This scenario presents three key integration challenges—reconciling cross-technology batch effects, aligning anatomical structures across developmental time points and harmonizing representations across drastically different spatial scales. Preliminary UMAP visualization confirmed strong batch effects across technologies and time points (Supplementary Fig. 5a). For benchmarking, we tested SpaMosaic, CLUE, Cobolt, MIDAS, StabMap, scMoMaT, MultiVI and UINMF.

SpaMosaic successfully harmonized the samples, generating spatially coherent domains that aligned consistently across developmental stages and technologies (Fig. 4d). Referencing the Allen Brain Atlas[28] (Supplementary Fig. 6a), we identified major brain regions, including the dorsal pallium (DPall), olfactory bulb, subpallium, hypothalamus, diencephalon, midbrain and hindbrain (detailed cluster assignments in Fig. 4d). Visualization of relevant markers confirmed their identities (Supplementary Fig. 6b). For example, SpaMosaic subdivided the DPall into two distinct clusters (DPallm and DPallv), with DPallm exhibiting higher expression of *Mef2c*[25] and *Tbr1* (ref. 43). In the subpallium, three transcriptionally distinct clusters were identified and validated by markers including *Foxp1* (cortical excitatory neurons[44]), *Dlx1* (GABAergic interneuron development[45]) and *Lhx8*/*Adarb2* (caudal ganglionic eminence-derived inhibitory interneurons[46]). Similarly, SpaMosaic resolved multiple clusters within the diencephalon, hindbrain and ventricular zones, confirmed by visualizing their respective markers.

To demonstrate SpaMosaic's multimodal information integration, we analyzed the Misar-seq samples by visualizing region-specific markers *Rarb* (subpallium-cortex), *Pax6* (DPallv) and *Neurod6* (DPallm[25,47]; Fig. 4e). These genes displayed distinct expression patterns across the RNA and ATAC modalities—*Rarb* showed enhanced RNA expression in the subpallium, while its ATAC-derived gene activity was uniformly distributed without clear spatial specificity. Conversely, *Pax6* and *Neurod6* showed more pronounced region-specific activity in the ATAC modality than in the RNA modality. Together, these results illustrate that SpaMosaic leverages complementary RNA and ATAC signals to achieve integrative spatial characterization.

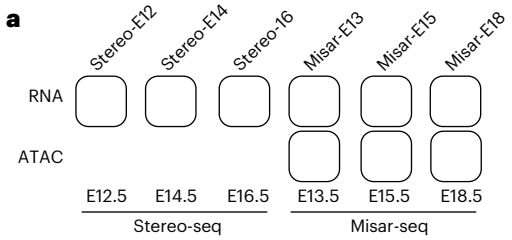
By contrast, most competing methods produced noisy clustering outputs that poorly matched known anatomy, particularly for the Misar-seq sections (Fig. 4d). MIDAS and MultiVI identified major brain regions in the Stereo-seq sections but showed limited or inconsistent structure recovery in the Misar-seq sections, indicating suboptimal cross-platform integration. CLUE failed to distinguish key regions like the DPallm from DPallv and merged distinct hindbrain and midbrain structures. Finally, Cobolt, StabMap, scMoMaT, UINMF, BANKSY and CellCharter produced noisy or inconsistent clusters across the two technologies' sections (Fig. 4d and Supplementary Fig. 7).

Quantitative benchmarking confirmed these observations. SpaMosaic and MultiVI achieved similar PAS and CHAOS scores that outperformed others (Fig. 4f), although SpaMosaic captured more anatomical brain structures as described above. For batch integration (iLISI), UINMF achieved the highest score, followed by Cobolt (Fig. 4g). However, UMAP visualizations showed that SpaMosaic maintained clear cluster boundaries with reasonable section mixing, whereas Cobolt produced diffused clusters with overlap (Supplementary Fig. 5b).

To further assess SpaMosaic's robustness across diverse platform configurations, we evaluated two additional mosaic scenarios with pronounced disparities in resolution and capture area. The first

Fig. 4 | Downstream analysis results obtained with SpaMosaic and quantitative benchmarking on an embryonic mouse brain (Misar + Stereo) dataset. a, Schematic of the dataset composition. Mouse embryonic brain sections at developmental stages E13.5, E15.5 and E18.5 were profiled for the RNA and ATAC modalities using Misar-seq, and sections at E12.5, E14.5 and E16.5 were profiled for RNA expression using Stereo-seq. **b**, Comparison of resolution and spot numbers between Stereo-seq (25 µm; 15,000 cells per section) and Misar-seq (50 µm; 2,000 cells per section). **c**, Spatial plots of modality-specific clustering results obtained by applying the Leiden algorithm to batch-corrected representations, jointly across relevant sections. **d**, Spatial plots of the clusters obtained from mosaic integration methods for the six sections. Manual annotation was performed to identify SpaMosaic's clusters. **e**, RNA expression and ATAC-derived gene activity maps for *Rarb*, *Pax6* and *Neurod6* in the three Misar-seq sections. **f**, PAS and CHAOS scores of benchmarked methods. Box plots display the distribution of scores calculated across the six sections ($n = 6$). Centerlines indicate the median; box limits represent the IQR; whiskers denote 1.5× IQR; dots indicate outliers beyond 1.5× IQR. **g**, iLISI scores of benchmarked methods. Bars represent the global score calculated for the integrated dataset ($n = 1$).

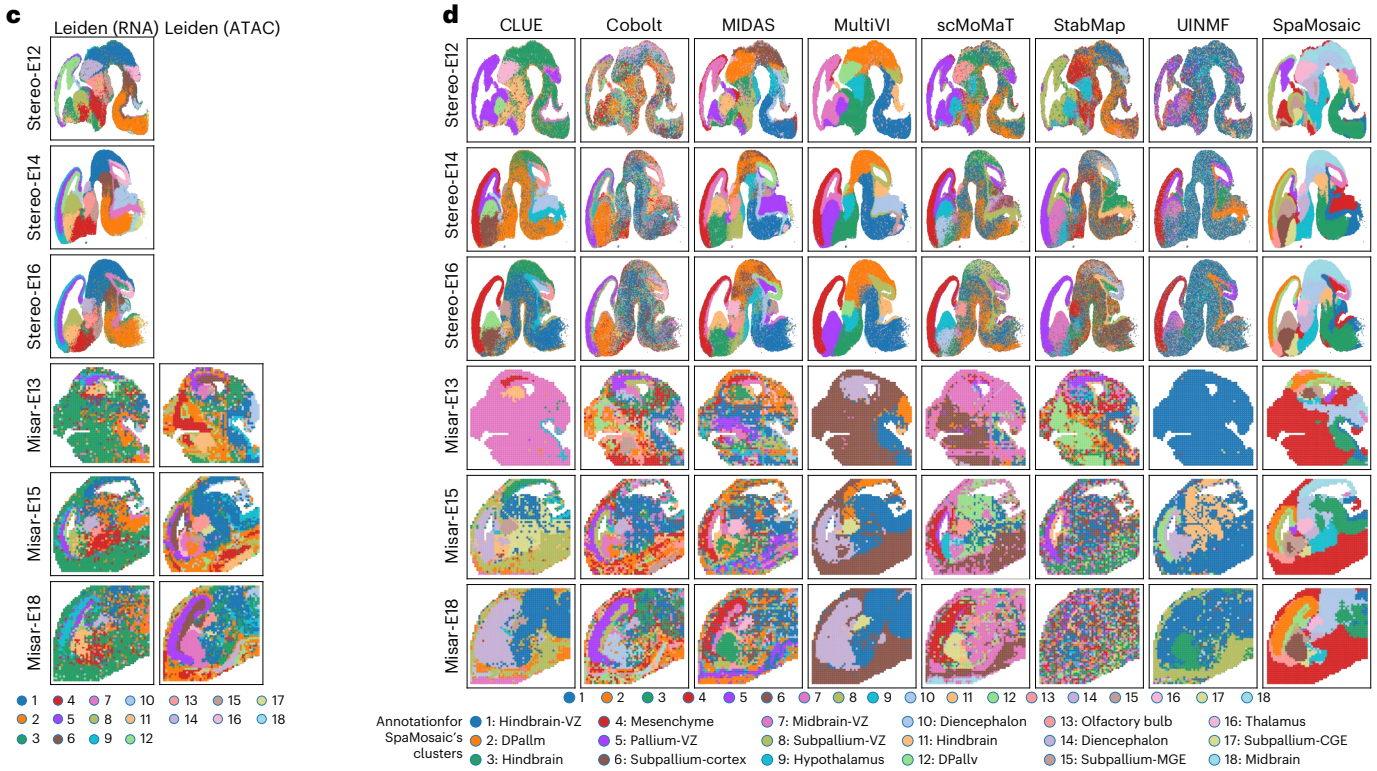
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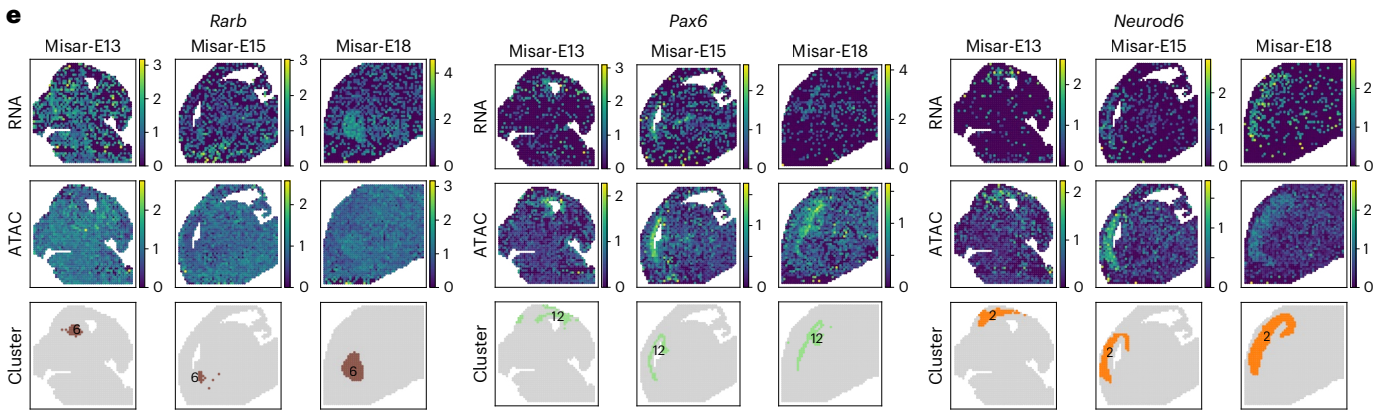


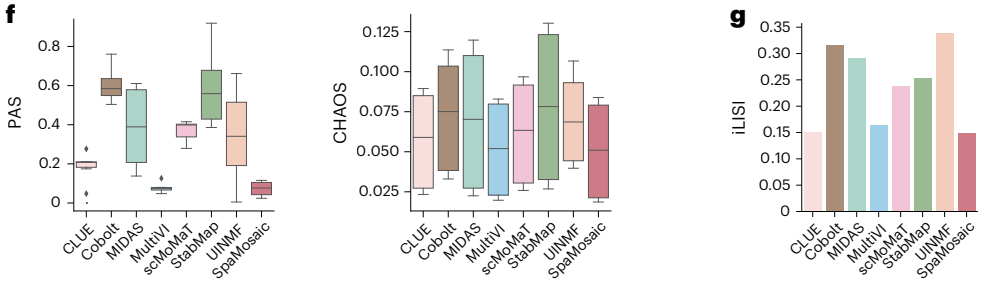
b

Embryonic mouse brain

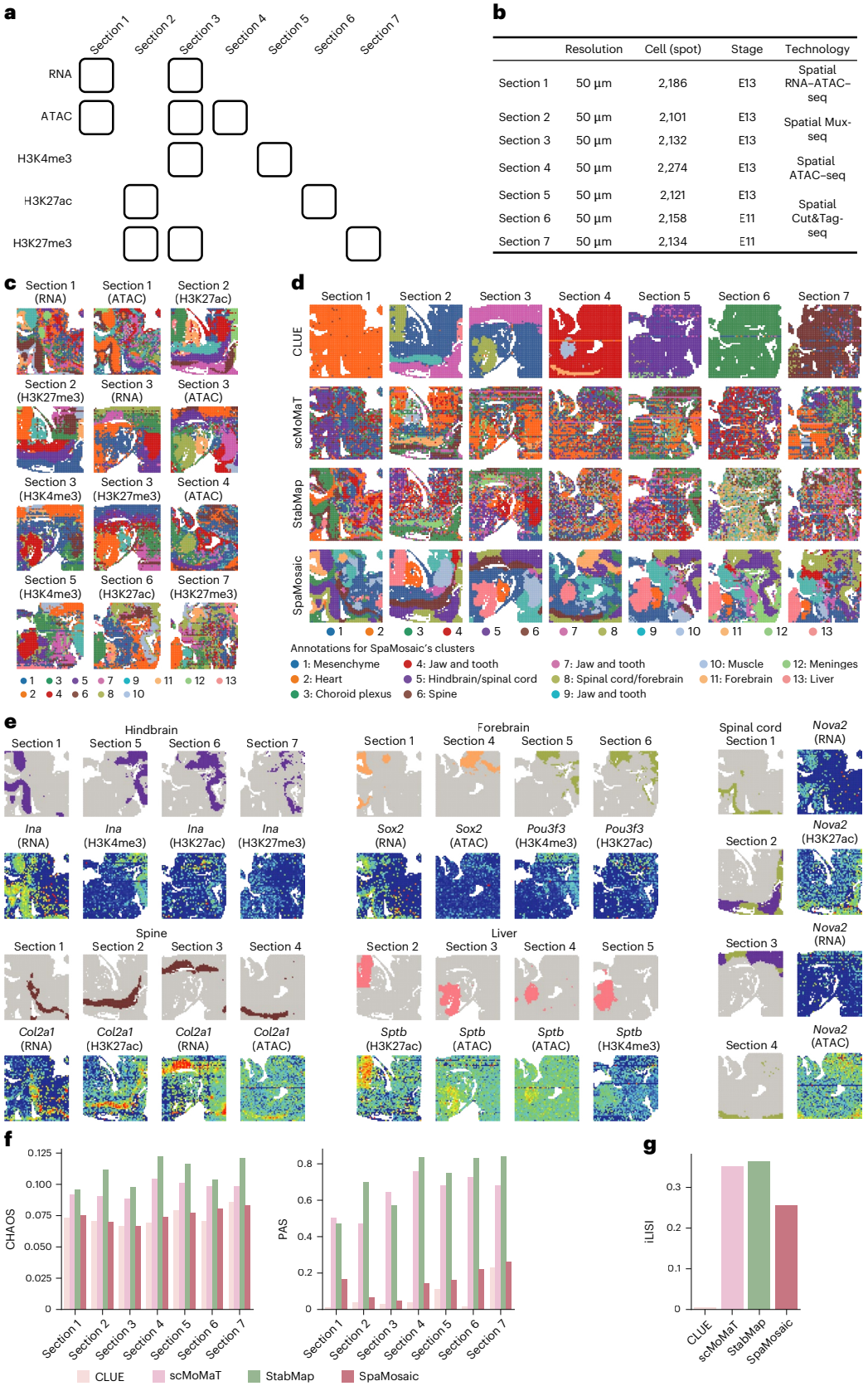
	Stereo-E12	Stereo-E14	Stereo-E16	Misar-E13	Misar-E15	Misar-E18
Resolution	25 μm	25 μm	25 μm	50 μm	50 μm	50 μm
Cell (spot)	9,556	17,594	17,329	1,777	1,949	2,129







https://doi.org/10.1038/s41588-026-02573-3



	Resolution	Cell (spot)	Stage	Technology
Section 1	50 μm	2,186	E13	Spatial RNA–ATAC–seq
Section 2	50 μm	2,101	E13	Spatial Mux-seq
Section 3	50 μm	2,132	E13	
Section 4	50 μm	2,274	E13	Spatial ATAC–seq
Section 5	50 μm	2,121	E13	Spatial Cut&Tag-seq
Section 6	50 μm	2,158	E11	
Section 7	50 μm	2,134	E11	

Fig. 5 | Application of SpaMosaic to mouse embryo data integration. a, Layout of input data matrices in the mouse embryo dataset. **b**, Metadata for each section in the mouse embryo dataset. **c**, Modality-specific clustering was performed jointly across relevant sections using the Leiden algorithm on batch-corrected representations. **d**, Spatial plots of clusters obtained from the methods for all seven sections. Manual annotation was performed to identify SpaMosaic's clusters. **e**, Spatial heatmap of RNA expression, GAS and CSS for region-specific markers in SpaMosaic's clusters. **f**, CHAOS and PAS scores of benchmarked methods. Bars represent the scores calculated individually for each of the seven tissue sections ($n = 1$). **g**, iLISI scores of benchmarked methods. Bars represent the global score calculated for the integrated dataset ($n = 1$).

integrated Visium, spatial Mux-seq[4] and Visium HD[48], spanning a greater than 30-fold imbalance in data density (~2,700 to ~99,000 spots) and resolutions ranging from 16 µm to 55 µm. The second was a postnatal mouse brain dataset of three different technologies (DBiT-seq[3]; spatial RNA–ATAC–seq, spRA[2]; spatial ATAC–seq, spA[49]) with varying field-of-view and spot counts. SpaMosaic consistently outperformed competitors in preserving spatial continuity and resolving fine anatomical structures, such as the hippocampus and cortical layers. Detailed benchmarking and validation are provided in Supplementary Note 4.

SpaMosaic facilitates integration of highly heterogeneous data for mouse embryo atlasing

To demonstrate SpaMosaic's efficacy at comprehensive spatial atlasing, we performed a proof-of-principle integration of mouse embryogenesis, spanning critical developmental stages (E11–E13). The dataset comprises seven tissue sections profiled with four technologies—spatial RNA–ATAC–seq, spatial Mux-seq, spatial ATAC–seq and spatial CUT&Tag-seq (Fig. 5a). Crucially, the molecular composition varies drastically across sections. Sections 1–3 profiled diverse multimodal mixtures—RNA and ATAC (section 1); H3K27ac and H3K27me3 (section 2); and RNA, ATAC, H3K4me3 and H3K27ac (section 3). Sections 4–7 consist exclusively of single-modality epigenomic profiles (ATAC or histone modifications). This scenario includes several core challenges in building a unified anatomical reference—the integration must overcome substantial technical and anatomical variations across developmental time points (Fig. 5b,c and Supplementary Fig. 8), while simultaneously resolving substantial differences in molecular coverage. Notably, this scenario's complexities rendered several single-cell mosaic integration methods (Cobolt, MIDAS, MultiVI, UINMF) inapplicable, limiting our comparisons to SpaMosaic, StabMap, scMoMaT and CLUE.

Among the tested methods, SpaMosaic demonstrated superior ability in identifying consistent spatial domains across all sections, despite differences in section orientation and modality composition (Fig. 5d). Marker gene expression patterns confirmed that SpaMosaic successfully recovered key tissues and organs (Fig. 5e). For instance, the hindbrain marker *Ina*[42] had high RNA expression (cluster 5, section 1), elevated H3K4me3 and H3K27ac GAS (cluster 5, sections 5 and 6) and reduced H3K27me3 CSS (cluster 5, section 7). The forebrain markers[42] *Sox2* and *Pou3f3* also showed high RNA expression (cluster 11, section 1) and increased ATAC or histone modification GAS (cluster 11 in section 4, cluster 8 in sections 5 and 6). Similarly, the markers for spine (*Col2a1* (ref. 2)), liver (*Sptb*[49]) and spinal cord (*Nova2* (ref. 50)) showed consistent expression or gene activity patterns across their corresponding clusters despite differences in the measured modalities. Finally, markers for the heart, jaw and teeth (mandible), mesenchyme, meninges, muscle and choroid plexus also displayed expected expression patterns in their respective clusters (Extended Data Fig. 4).

In contrast, the competing methods struggled to recover consistent anatomical structures. CLUE failed to identify spatial domains in sections 1, 5, 6 and 7, and captured limited anatomical features in others. scMoMaT and StabMap produced spatially discontinuous clusters with few identifiable anatomical structures. Single-modality spatial clustering algorithms (BANKSY and CellCharter) identified more coherent clusters but showed incomplete anatomical coverage and inconsistent cluster assignments across sections, and had poor section mixing in the UMAP visualizations (Supplementary Fig. 9).

The quantitative metrics matched the observations—SpaMosaic achieved the top CHAOS scores on sections 2, 3, 5 and 7, and ranked second at PAS across all sections (Fig. 5f). While CLUE scored highest on spatial coherence metrics, its domains showed limited anatomical correspondence as described above. For batch integration (iLISI), SpaMosaic ranked third after StabMap and scMoMaT (Fig. 5g). CLUE's near-zero iLISI score and poor section mixing in UMAP visualizations (Supplementary Fig. 10) contrasted with its strong spatial coherence scores. Overall, this scenario illustrates that SpaMosaic can effectively integrate disparate data to capture spatially coherent domains with biological relevance.

Discussion

Spatial omics technologies provide powerful means to interrogate tissue sections across different molecular layers. By measuring different omes, we obtain complementary views of the tissue that, coupled with spatial information, enable deeper insights into underlying biology. However, technological limitations and high costs hinder the acquisition of multiple high-quality omics profiles from a single section. An alternative approach is to integrate data acquired from different omes across different sections. In this work, we presented SpaMosaic for such spatial mosaic integration. It leverages a contrastive learning framework to construct modality-agnostic latent spaces and uses GNNs to encode spatial proximity and omics feature similarity, achieving spatially aware integration. The contrastive learning framework bridges the information gap between modalities, which has been validated as a simple but effective strategy for modality alignment and semantic preservation in various applications[51–54]. Moreover, it is also highly flexible and allows easy scaling to include more data modalities, as demonstrated in our five-modal mosaic integration example.

SpaMosaic demonstrated accurate and robust consensus spatial clustering, outperforming other methods across different tissue types and technology platforms. By integrating information from different sections and omics modalities, SpaMosaic reduces noise to enhance spatial clustering results. SpaMosaic is not restricted to integrating data from adjacent sections; it successfully integrates data from different time points, studies and technology platforms. Alongside cross-modality integration, it simultaneously achieves cross-batch integration to remove technical effects while preserving biological variations. This enables identification of consensus cell types and tissue structures across samples, as well as capture of subtle differences among sample groups. In addition to data integration, SpaMosaic imputes missing molecular layers within mosaic datasets using a *k*-nearest neighbors (*k*NN)-based approach that leverages the modality-aligned latent space, enhancing interpretability.

SpaMosaic can be further enhanced in several ways. First, improving scalability to handle larger datasets with more tissue sections will be critical for constructing spatial multi-omics atlases. Techniques such as graph sampling and caching of intermediate outputs can improve computational performance[55]. Second, incorporating continual learning[56] and transfer learning[57] approaches would enable analysis with new sections without retraining, and also offline analysis of private datasets with pretrained models to address data privacy concerns. Finally, incorporating accompanying image data, such as staining images, as an additional modality could further enhance accuracy in cell type and tissue domain identification.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-026-02573-3.

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Methods

Data preprocessing

For each spot (cell) $i \in \{1, \ldots, n_b\}$ from section $b \in \{1, \ldots, B\}$, let $c_i^m \in R^{d^m}$ of size d^m be the omics feature count vector of spot i from modality $m \in U = \{m_1, \ldots, m_M\}$ where U is the set of all RNA, epigenomic and protein modalities. For each modality, we first normalized the count matrix using log-normalization for RNA, TF-IDF transformation for epigenomic modality and centered log ratio normalization for protein. Thereafter, we selected the top 5,000 highly variable features for RNA and top 100,000 for epigenome, followed by singular value decomposition into low-dimensional representations. To correct batch effects within a modality, we used Harmony or Seurat to integrate these low-dimensional representations. For spot i with modality m, the batch-corrected representation was x_i^m, which was then used as input for model training.

Construction of the spot–spot graph

We encode both spatial and omics feature similarity in each modality with a spot–spot graph containing two edge types—spatial-based edges connecting spots within one section and omics feature-based edges connecting spots across sections. For spatial-based edges, we compute Euclidean distances across spots using their spatial coordinates and then identify each spot's kNN using Annoy (https://github.com/spotify/annoy). Finally, we construct an undirected spatial neighbor graph of all spots in modality m (all modalities share the same spatial neighbor graph) in the form of an adjacency matrix $A^{\text{intra}-m}$, where $A_{ij}^{\text{intra}-m} = 1$ if spots i and j are spatial neighbors.

For the omics feature-based edges, we use the batch-corrected low-dimensional representation to find the MNNs[21] between sections. Specifically, for each spot i in section b_1, we find its k_1 nearest neighbors in section b_2; for each spot j in b_2, we find its k_2 nearest neighbors in b_1. If spots i and j are in each other's nearest neighbors set, then (i, j) form a pair of MNNs. We then construct a neighbor graph for each modality m denoted by an adjacency matrix $A^{\text{inter}-m}$, where $A_{ij}^{\text{inter}-m} = 1$ if i and j belong to a pair of MNNs. For each modality m, we now combine the $A^{\text{intra}-m}$ and $A^{\text{inter}-m}$ into one graph:

$$A^m = A^{\text{intra}-m} + w_g A^{\text{inter}-m}$$

where w_g is a positive constant, denoting the weight for $A^{\text{inter}-m}$. The contribution of each graph component was evaluated through ablation studies (Supplementary Note 5), and we assessed the robustness of SpaMosaic to the choice of k (neighborhood size) in the spatial adjacency graph construction (Supplementary Note 6).

When section spot counts are highly imbalanced, using equal neighbor counts ($k_1 = k_2$) tends to bias the MNN set. Specifically, most spots from the smaller section are selected, while only a small fraction of spots from the larger section are used. This is analogous to scaling up the small section to match the large one, although the smaller section may only occupy a fraction of the larger section's area. To mitigate this, we adopt asymmetric neighbor sizes when section sizes are substantially imbalanced. For a base neighbor count k_0, if the imbalance exceeds a threshold θ (that is, $n_1/n_2 < \theta$, where n_1 and n_2 represent the spot counts of the two sections), the smaller section uses k_0, and the larger section uses $\max(1, \lfloor k_0 \times (n_1/n_2) \rfloor)$. Otherwise, both sections use k_0. To further prune potential outliers, we apply an isolation forest[58] to remove 50% MNN pairs with the highest anomaly scores. The input features to the isolation forest model consist of spatial offsets (absolute coordinate differences across paired spots) concatenated with their coordinates. Both the asymmetric neighbor sizing and outlier removal steps are optional within our pipeline.

Graph embedding

The spot–spot graph encodes information about the spatial proximity and omics feature similarity across spots, which we integrate with the omics profiles into spot embeddings through GNNs. However, our spot–spot graph contains two types of edges, each with different semantics. This differs from the traditional homogeneous graphs and, thus, we explored various approaches to address this situation, namely the graph attention network[59,60], heterogeneous graph transformer[61] and two variants of light graph convolution network (LGCN[62,63]; Supplementary Note 7). In our final model, we adopt the WLGCN, which is defined as:

$$\tilde{X}^m = \text{LGCN}^m(A^m, X^m) = \text{Concat}\left(X^m, \hat{A}^m X^m, \hat{A}^{m^2} X^m, \ldots \hat{A}^{m^l} X^m\right)$$

where m denotes the modality and l denotes the number of layers. $\hat{A}^m = \tilde{D}^{m^{-\frac{1}{2}}}(A^m + I)\tilde{D}^{m^{-\frac{1}{2}}}$, $\tilde{D}^m$ is the diagonal degree matrix of $A^m + I$. The embeddings are then transformed by a multiple layer perceptron (MLP):

$$Z^m = f_{\text{mlp}}^m\left(\tilde{X}^m\right)$$

where m denotes the modality. Finally, the graph embedding function is defined as g:

$$Z^m = g^m(A^m, X^m) = f_{\text{mlp}}^m\left(\text{LGCN}^m(A^m, X^m)\right)$$

By setting w_g, we make A^m a weighted graph to control the importance between spatial proximity and omics feature similarity. From our testing, we decided on $w_g = 0.8$ for our benchmarking examples, giving more weight to spatial proximity than omics feature similarity, as we consider the former to be more reliable due to measurement noise present in the omics profiles.

Contrastive learning

Within SpaMosaic, the model's goal is to learn a set of modality-agnostic low-dimensional embeddings, thereby aligning multiple sections with different modalities. SpaMosaic uses contrastive learning to leverage the input data's multimodal nature to achieve this goal. In the contrastive learning framework, two views of one sample constitute a positive pair, while views from different samples constitute negative pairs[18]. By pushing positive pairs together while separating negative pairs in the latent space, contrastive learning achieves alignment across the different views and preserves variation across samples. For SpaMosaic, two modalities of one spot constitute a positive pair and modalities from different spots constitute negative pairs. Thus, for two modalities m_1 and m_2, there is a sampled mini-batch of spots $\{x_1^{m_1}, \ldots, x_n^{m_1}\}$ and $\{x_1^{m_2}, \ldots, x_n^{m_2}\}$ that form positive pairs and the associated contrastive learning objective is:

$$L_{\text{con}}^{m_1,m_2} = -\frac{1}{2n}\sum_{i=1}^{n}\left[\log\left(\frac{\exp(s_{i,i}^{1,2})}{\sum_{l=1}^{n}\exp(s_{i,l}^{1,2}) + \sum_{l\neq i}\exp(s_{i,l}^{1,1})}\right) + \log\left(\frac{\exp(s_{i,i}^{2,1})}{\sum_{l=1}^{n}\exp(s_{i,l}^{2,1}) + \sum_{l\neq i}\exp(s_{i,l}^{2,2})}\right)\right] \quad (1)$$

where

$$s_{i,l}^{1,2} = \frac{1}{\tau} \times \frac{\left\langle z_i^{m_1}, z_l^{m_2}\right\rangle}{||z_i^{m_1}|| \times ||z_l^{m_2}||} = \frac{1}{\tau} \times \frac{\left\langle g^{m_1}\left(x_i^{m_1}\right), g^{m_2}\left(x_l^{m_2}\right)\right\rangle}{||g^{m_1}\left(x_i^{m_1}\right)|| \times ||g^{m_2}\left(x_l^{m_2}\right)||},$$

g^{m_1}, g^{m_2} denote the modality-specific graph encoder network, $z_i^{m_1}, z_l^{m_2}$ denote the modality-specific embeddings, $\langle \cdot, \cdot \rangle$ denotes inner product of two vectors and τ is a positive constant. The numerator in the contrastive learning objective aims to bring closer the embeddings of each spot's two modalities, while the first term in the denominator aims to separate the embeddings of two modalities from different spots and the second term aims to separate the embeddings within the same modality but from different spots.

Combining embeddings from the measured modalities, the unique representation for spot l in section b is defined as:

$$z_l = \sum_{m \in T_b} w_m z_l^m$$

where T_b denotes the set of modalities that section b is measured with, $w_m \in (0, 1)$ denotes the weight for modality m and $\sum_{m \in T_b} w_m = 1$.

Feature reconstruction

Based on our definition, contrastive learning can only be applied to spots with multimodal profiles. For the remaining spots without multimodal profiles, we add a feature reconstruction objective to include them in the model training. We also use a modality-specific MLP decoder, r^m, to reconstruct the inputs x_i^m based on the embeddings z_i^m. The reconstruction loss is defined as:

$$L_{\text{rec}}^m = \frac{1}{n} \sum_{i=1}^{n} ||r^m(z_i^m) - x_i^m||_2^2$$

Thus, the final loss function for the whole model is defined as:

$$L = \sum_{m_i, m_j \in U} L_{\text{con}}^{m_i, m_j} + \sum_{m \in U} L_{\text{rec}}^m$$

where $U = \{m_1, \ldots, m_M\}$.

Implementation details

For each modality, the graph encoder network comprises two WLGCN layers, followed by a fully connected layer with a dimension of 128, leaky ReLU activation (slope = 0.2)[64], batch normalization[65], dropout ($P = 0.2$)[66] and a final fully connected layer with a dimension of 32. The decoder comprises one fully connected layer with linear activation. In our implementation, the GNNs use the PyTorch Geometric library[67]. To train the model, we used the Adam optimizer[68] with a learning rate of 0.01. The weight decay was set to 5×10^{-4}, the number of training epochs was set to 100 and the temperature parameter τ was set to 0.01. The parameter k used to calculate the spatial adjacency graph was set to 10 across all experiments. For MNN calculation, we used $k_0 = 10$ across all experiments. When asymmetric neighbor sizing was applied, the imbalance threshold was set to $\theta = 0.8$. For outlier removal with isolation forest, we used scikit-learn's implementation with the default parameters[69]. Our SpaMosaic implementation supports both GPU and CPU execution, with GPU acceleration recommended for improved computational efficiency.

Imputation

Let $Z^{m_j} = \{z_1^{m_j}, \ldots, z_n^{m_j}\}$ denote the graph embeddings for all spots that were measured in modality m_j. Alongside, $C^{m_j} = \{c_1^{m_j}, \ldots, c_n^{m_j}\}$ denotes the feature profiles (counts, GAS, etc.) for all spots that were measured with modality m_j. Let T_b denote the set of modalities that section b was measured with and $\overline{T}_b = U - T_b$ denotes the set of modalities that are missing in section b. To impute the missing modality $m_j \in \overline{T}_b$ for b, we first perform cross-modal matching between Z^{m_j} and $\{z^{m_t} | m_t \in T_b\}$. Specifically, for each $z_l^{m_t}$ in section b, we find the indices of its k_2 nearest neighbors from Z^{m_j}, $Q = \{q_1, q_2, \ldots, q_{k_2}\}$. The profiles $c_l^{m_j}$ imputed from modality m_t are thus defined as:

$$c_l^{m_t \to m_j} = \frac{1}{k_2} \sum_{q \in Q} c_q^{m_j}$$

where $m_t \to m_j$ denotes the profiles imputed based on modality m_j. For each modality in T_b, we repeat the above imputation process and the final imputed profiles for m_j in section b are given by:

$$c_l^{m_j} = \frac{1}{|T_b|} \sum_{m_t \in T_b} c_l^{m_t \to m_j}$$

where $|T_b|$ denotes the number of elements in T_b.

Data simulation

Following the procedure discussed in ref. 22, we simulated multivariate count data with the 'ggblocks' spatial pattern. The ground-truth spatial factors comprised four simple shapes in different spatial regions. For each section created, the spots were arranged in a 36×36 grid for 1,296 spots. The number of RNA features was set to $J^{rna} = 200$, and the number of protein features was set to $J^{\text{pro}} = 60$. Each feature was randomly assigned to one of the four factors with uniform probability. For the RNA modality, its count matrix was drawn from a zero-inflated negative binomial distribution with a mean $M^{\text{RNA}} = b^{\text{RNA}} + S^{\text{RNA}} F^{\text{RNA}} W^{\text{RNA}}$ where $S^{\text{RNA}} \in R^{1,296 \times 1,296}$, $F^{\text{RNA}} \in R^{1,296 \times 4}$ and $W^{\text{RNA}} \in R^{4 \times 200}$. $F_{il}^{\text{RNA}} = 1$ if spatial factor l was active in spatial location i and $F_{il}^{\text{RNA}} = 0$ otherwise. By default, $W_{jl}^{\text{RNA}} = 10$ if feature j was assigned to spatial factor l and $W_{jl}^{\text{RNA}} = 0$ otherwise. Finally, b^{RNA} represents the background mean expression value. To simulate a spatial factor l mixing with the background factor, we multiplied F^{RNA} by a diagonal matrix S^{RNA}. For those entries corresponding to factor 1 we set them to s^{RNA}, while for factors 2 and 3, we randomly selected 40% of entries to be assigned s^{RNA} and the rest were set to 1. The shape parameter for the negative binomial component was set to 20 and the dropout probability was set to p^{RNA}. For the protein modality, its count matrix was drawn from a negative binomial distribution with a mean $M^{\text{pro}} = b^{\text{pro}} + S^{\text{pro}} F^{\text{pro}} W^{\text{pro}}$, where $S^{\text{pro}} \in R^{1,296 \times 1,296}$, $F^{\text{pro}} \in R^{1,296 \times 4}$ and $W^{\text{pro}} \in R^{4 \times 60}$. $F_{il}^{\text{pro}} = 1$ if spatial factor l was active in location i and 0 otherwise. By default, $W_{jl}^{\text{pro}} = 12$ if feature j was assigned to pattern l and $W_{jl}^{\text{pro}} = 0$ otherwise. S^{pro} is a diagonal matrix and for those entries corresponding to factor 4 we set them to s^{pro}. For factors 2 and 3, we randomly selected 40% of entries to be assigned s^{pro}, and the other entries were set to 1. The shape parameter was set to 10.

To simulate batch effects between sections, we changed the W^{rna} and W^{pro} for the second and the third sections, respectively. Specifically, if the RNA feature set $\{j_1^{\text{RNA}}, j_2^{\text{RNA}}, \ldots j_{n_l}^{\text{RNA}}\}$ was assigned to spatial factor l, we randomly picked half of these features and increased each W_{jl}^{RNA} by a factor of 3. Similarly, if protein feature set $\{j_1^{\text{RNA}}, j_2^{\text{RNA}}, \ldots j_{v_l}^{\text{pro}}\}$ was assigned to spatial factor l, we randomly picked half of these features and increased each corresponding W_{jl}^{pro} by a factor of 1.1, while for the remaining features, we increased each W_{jl}^{pro} by a factor of 3.

We designed five sets of simulation parameters (Extended Data Fig. 1a) and created three datasets with different random seeds for each parameter set. Each dataset consisted of three sections sharing the same simulation parameters. The first section contained RNA and protein expression, the second section contained only RNA expression and the third section contained only protein expression (Extended Data Fig. 1b).

Benchmarking metrics for integration

We quantitatively evaluated SpaMosaic across four aspects—spatial domain identification, continuity of identified spatial domains, batch integration and modality alignment. For spatial domain identification and continuity, we used the ARI, PAS (percentage of spots differing from ≥6 of their ten spatial neighbors) and CHAOS (average minimal intra-cluster spatial distance)[23]. Batch integration was assessed using graph iLISI[24], which ranges from 0 (strong batch separation) to 1 (perfect batch mixing). Modality alignment was evaluated using fraction of samples closer than true match[70], matching score[71] (based on MNNs) and label transfer F1 score[10]. The detailed metric definitions and formulas are provided in Supplementary Note 8.

Benchmarking methods for mosaic integration

In our benchmarking, we tested SpaMosaic and other state-of-the-art single-cell mosaic integration methods, CLUE, Cobolt, scMoMaT, StabMap, MIDAS, UINMF and MultiVI. We also tested a nonspatial version of SpaMosaic. Here we describe the detailed settings used for each method.

CLUE. We used CLUE from the scglue package (version 0.1.1)[72] in Python. We followed the processing pipeline as described by the authors (https://github.com/openproblems-bio/neurips2021_multimodal_topmethods/tree/main/src/match_modality/methods/clue). Raw count matrices of each modality were used as input, which were then transformed by CLUE's customized preprocessing steps. Thereafter, we applied the default model training pipeline. We set shared_batches = False and used the recommended training parameters for different types of input data, namely RNA + ATAC and RNA + protein.

Cobolt. Although originally designed for RNA and ATAC–seq integration, Cobolt's statistical framework is generalizable to other omics modalities through its multinomial modeling approach[8]. We ran the Cobolt package (version 1.0.1) following its tutorial (https://github.com/epurdom/cobolt/blob/master/docs/tutorial.ipynb). We first selected the highly variable features (top 5,000 for RNA, top 100,000 for epigenome) from the count data matrices before inputting them to the model for training. Most of the training hyperparameters were set to their default values—training iteration = 100, latent dimension = 10, batch size = 128. However, we modified the learning rate from 0.005 to 0.001 to avoid the numerical issues encountered with our datasets. When running the 'calc_all_latent' function for the postnatal mouse brain dataset (spRA + spRH), we set the parameters target = [True, False, False, False, False] for successful execution.

scMoMaT. We used the Python package scMoMaT (version 0.2.2) by following its tutorial (https://github.com/PeterZZQ/scMoMaT/blob/main/demo_scmomat.ipynb). We first selected the highly variable features (top 5,000 for RNA, top 100,000 for epigenome) in the count data matrices as input for model training. The model parameters were kept at their defaults, except that we disabled the generation of pseudocount RNA matrix from ATAC data for better performance with our examples.

StabMap. We used the StabMap package in R (version 1.3.0), following its tutorial (https://www.bioconductor.org/packages/devel/bioc/vignettes/StabMap/inst/doc/stabMap_PBMC_Multiome.html). We first performed data normalization using the 'logNormCounts' function for all modalities. Thereafter, we selected the highly variable features using the 'modelGeneVar' function with modality-specific filtering thresholds (mean > 0.01 and $P \leq 0.05$ for RNA and protein, mean > 0.05 and $P < 0.1$ for epigenomic data). The model was trained using the default parameters for all runs.

MIDAS. We ran the MIDAS package following its tutorial (https://sc-midas-docs.readthedocs.io/en/latest/mosaic.html). The highly variable features (top 5,000 for RNA, top 100,000 for the epigenome) were first selected, and the resulting count matrices were used for training. For epigenomic input, the features were split by chromosome identifier (that is, 1–22, X and Y) to reduce model complexity. For all runs, the number of training epochs was set to 2,000.

UINMF. We ran the UINMF package (version 1.0.1) following the official tutorials (https://github.com/welch-lab/liger/tree/v1.0.1/vignettes) and a benchmarking tutorial for single-cell multi-omics integration methods (https://github.com/QuKunLab/MultiomeBenchmarking/tree/main/code/Integration/pipeline/Mosaic). Specifically, we first normalized the raw count matrices using the 'normalize' function for all modalities. Then, we selected highly variable features using the 'selectGenes' function with modality-specific variance thresholds (var.thresh = 0.00001 for epigenomic data and var.thresh = 0.1 for RNA data). Data scaling was then performed without zero-centering, and the scaled matrices were subjected to matrix factorization with a fixed dimensionality of 30 factors across all analyses.

MultiVI. We ran MultiVI by following the pipeline described in its tutorial (scVI version 0.19.0; https://docs.scvi-tools.org/en/stable/tutorials/notebooks/multimodal/MultiVI_tutorial.html). Raw count matrices were used as input and the training epoch was set to 100 while the other parameters were left at their default values.

SpaMosaic (nonspatial). SpaMosaic (nonspatial) is a modified version of SpaMosaic that uses an MLP as the encoder network without considering spatial information. The MLP encoder consists of two fully connected layers where the first layer is followed by a dropout layer ($P = 0.2$) and nonlinear activation exponential linear units[73]. The rest of the model components remain unmodified, and we ran it with the same settings as those for SpaMosaic.

CellCharter. We ran the CellCharter package (version 0.3.5) following its official tutorial (https://cellcharter.readthedocs.io/en/latest/). For RNA-seq data, dimensionality reduction and batch effect correction were performed using scVI with the zero-inflated negative binomial likelihood. ATAC–seq data were processed with scVI using the Poisson likelihood, while protein data were analyzed using a modified CellCharter-scArches[74] framework, in which the default negative binomial loss was replaced with a mean squared error loss. The number of nearest neighbors used to construct spatial neighborhoods was determined by the spatial technology. For instance, the Visium and Misar-seq acquired sections have regular spatial layouts, and therefore, we set the number of nearest neighbors to 6 and 8, respectively.

BANKSY. We ran the BANKSY package (version 1.2.1) following the official Python tutorial (https://github.com/prabhakarlab/Banksy_py/blob/Bansky_1.2.1/DLPFC_harmony_multisample.ipynb). We first selected for the top 2,000 highly variable genes. Principal component analysis was then performed on the BANKSY-generated spatially smoothed embeddings from individual tissue sections, followed by Harmony-based batch correction to integrate the dataset.

Clustering

For SpaMosaic, we applied model-based clustering using the MClust package[75] across all tests, with the parameter modelNames set to 'EEE'. For the other methods, we adopted the recommended clustering algorithms in their original publication or official tutorials. Specifically, for UINMF, Cobolt and MIDAS, the Louvain[76] clustering algorithm was used. For BANKSY, CLUE, MultiVI, scMoMaT and StabMap, Leiden clustering was applied. For SpaMosaic (nonspatial), MClust was applied. For CellCharter, we used its built-in Gaussian mixture model-based clustering approach as implemented in the official repository with all parameters set at their defaults. The impact of clustering method choice on benchmarking results is presented in Supplementary Note 9.

To determine the target number of clusters for each dataset, we first consulted the original publications or data portals to identify the number of clusters in each section. For each dataset, we selected the maximum number of clusters across all sections as the target number of clusters. The only exception was the postnatal mouse brain (DBiT + spRA + spA) dataset analyzed in Supplementary Note 4 where the four sections differ substantially in their coverage of anatomical regions. In this case, we used the median number of clusters across the four sections as the final target. To achieve the target number of clusters with Leiden and Louvain algorithms, we performed clustering at different resolutions ranging from 0 to 2 (with a 0.01 step size) and selected the resolution that yielded the cluster count closest to the target value.

Differential expression analysis and GO enrichment analysis

We used the Wilcoxon test, as implemented in the rank_genes_groups function of the SCANPY package (version 1.9.6)[77], to identify differentially expressed genes of spatial clusters. We performed the

GO enrichment analysis on cluster-specific differentially expressed genes using the R package clusterProfiler (version 4.6.2)[78].

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets used in this study were obtained from public repositories, except for the in-house generated human lymph node and tonsil datasets. The Misar-seq sections were downloaded from https://www.biosino.org/node/project/detail/OEP003285 and https://pan.baidu.com/s/15i0R5na2Jud_7EpkWSaZpA?pwd=oipj. The Stereo-seq sections were obtained from https://db.cngb.org/stomics/mosta/download/. The spatial RNA–Epigenomics sections were accessed through https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205055 and https://cells.ucsc.edu/?ds=brain-spatial-omics. The spatial ATAC–seq, spatial CUT&Tag-seq, spatial Mux-seq and DBiT–ARP-seq sections were retrieved from NCBI GEO under the accessions GSE171943, GSE165217, GSE263333 and GSE308623, respectively. The mouse brain 10x Visium and Visium HD datasets were obtained from the publicly available 10x Genomics resources (https://www.10xgenomics.com/datasets/mouse-brain-section-coronal-1-standard and https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-libraries-of-mouse-brain-he). All processed datasets are publicly available at https://zenodo.org/records/18946723 (ref. 79).

We used the scRNA-seq atlas of the mouse nervous system (available at http://mousebrain.org/adolescent/downloads.html) to assist in the analysis of cell types.

Code availability

The SpaMosaic open-source Python package is available at https://github.com/JinmiaoChenLab/SpaMosaic/tree/main, and all code for reproducing the benchmarking results presented in this study can be accessed at https://github.com/JinmiaoChenLab/SpaMosaic/tree/SpaMosaic-reproduce and archived on Zenodo (https://doi.org/10.5281/zenodo.18402871 (ref. 80)).

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Acknowledgements

The research is supported by Ministry of Education Academic Research Fund Tier 2 (CELL2VIRUS: AI-Driven Mapping of Virus-Host Interactions at Single-Cell and Spatial Resolution, MOE-T2EP30125-0005); NMRC Open Fund Large Collaborative Grant (Singapore lYMPHoma translatiONal studY (SYMPHONY) 2.0, MOH-001575-00); A*STAR under its BMRC Central Research Fund (UIBR) award; AI, Analytics and Informatics (AI3) Horizontal Technology Program Office seed grant (spatial transcriptomics in conjunction with GNNs for cell–cell interaction, C211118015) from A*STAR, Singapore; Open Fund Individual Research Grant (mapping hematopoietic lineages of healthy and high-risk acute myeloid leukemia patients with FLT3-ITD mutations using single-cell omics, OFIRG18nov-0103) from the Ministry of Health, Singapore; the National Research Foundation (NRF; award NRF-CRP26-2021-0001); the NRF, Singapore, and Singapore Ministry of Health's National Medical Research Council under its Open Fund-Large Collaborative Grant ('OF-LCG'; MOH-OFLCG18May-0003); and Singapore National Medical Research Council (NMRC/OFLCG/003/2018). M.L. is supported in part by the National Natural Science Foundation of China (grant 62225209 to M.L.). The authors thank L.A.R. Rodríguez-Kirby and G. Castelo-Branco from the Laboratory of

Molecular Neurobiology, Department of Medical Biochemistry and Biophysics, Karolinska Institute, Stockholm, Sweden, for generously sharing the DBiT–ARP-seq data. X.Y. and Z.F. conducted this work in part during an internship at the Bioinformatics Institute (BII), A*STAR, Singapore.

Author contributions

J.C. and M.L. initiated the project and provided funding support. J.C. conceived the idea. X.Y., J.C., Z.F. and R.Z. developed the method and designed the experiments. X.Y., Z.F., J.C. and K.S.A. performed the data analysis and wrote the manuscript. L.V.O., A.E., T.W. and D.G. provided the lymph node and tonsil datasets, and manually annotated the tissue structures. D.Z. and R.F. provided the postnatal mouse brain and mouse embryo datasets.

Competing interests

All authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41588-026-02573-3.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41588-026-02573-3.

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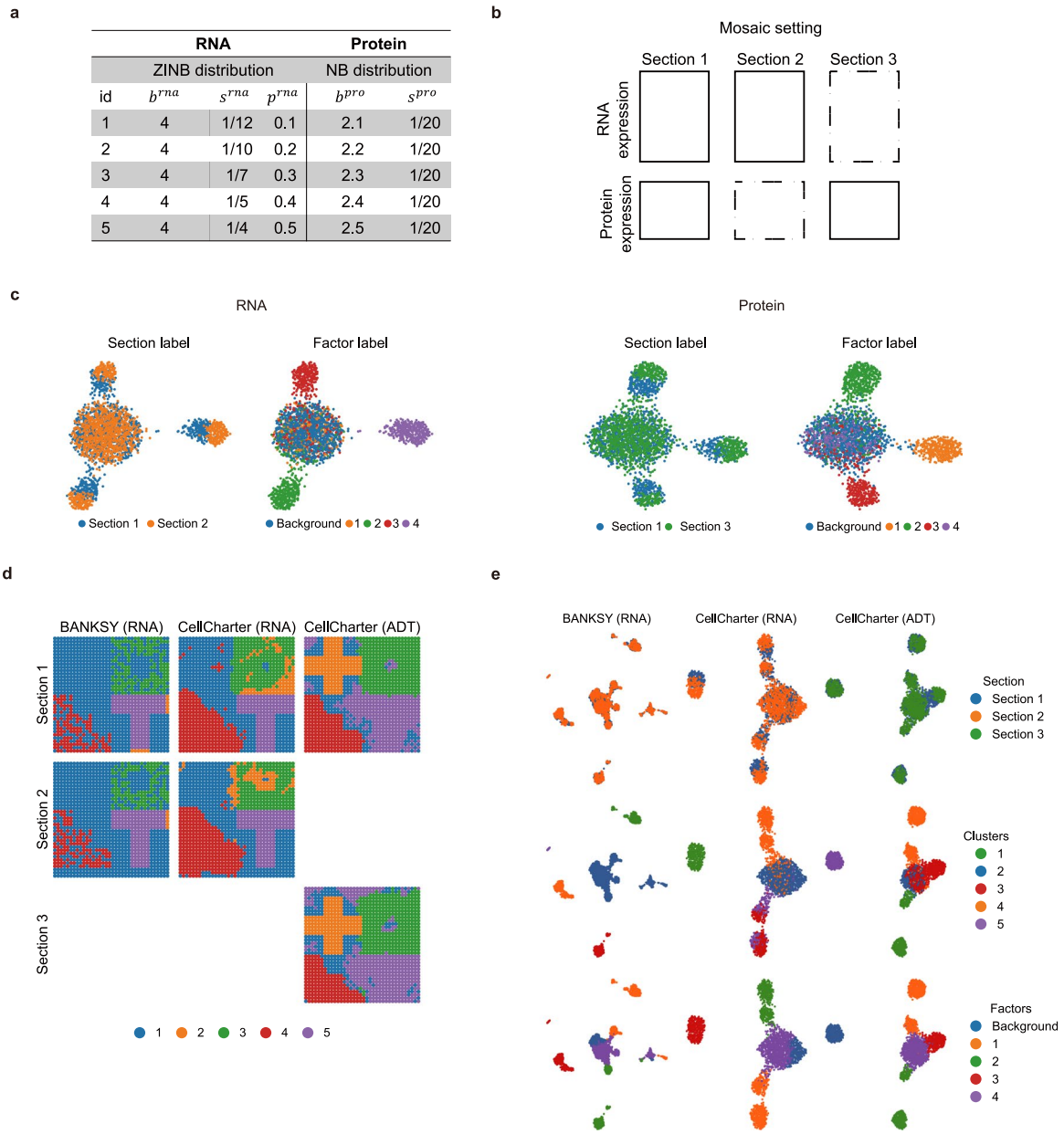
Peer review information *Nature Genetics* thanks Naveed Ishaque, Niklas Müller-Bötticher, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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https://doi.org/10.1038/s41588-026-02573-3

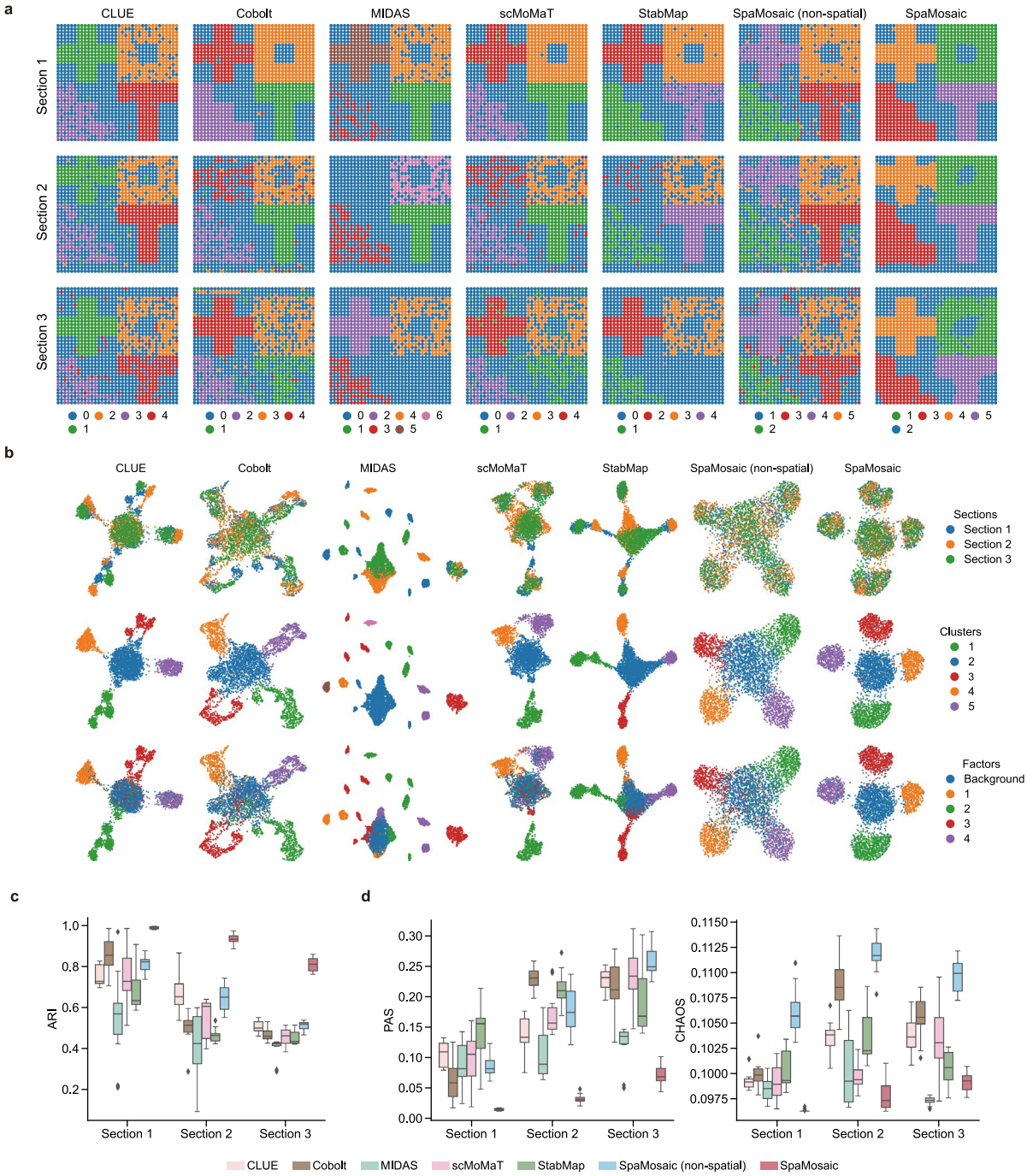
a

	RNA			Protein	
	ZINB distribution			NB distribution	
id	b^{rna}	s^{rna}	p^{rna}	b^{pro}	s^{pro}
1	4	1/12	0.1	2.1	1/20
2	4	1/10	0.2	2.2	1/20
3	4	1/7	0.3	2.3	1/20
4	4	1/5	0.4	2.4	1/20
5	4	1/4	0.5	2.5	1/20



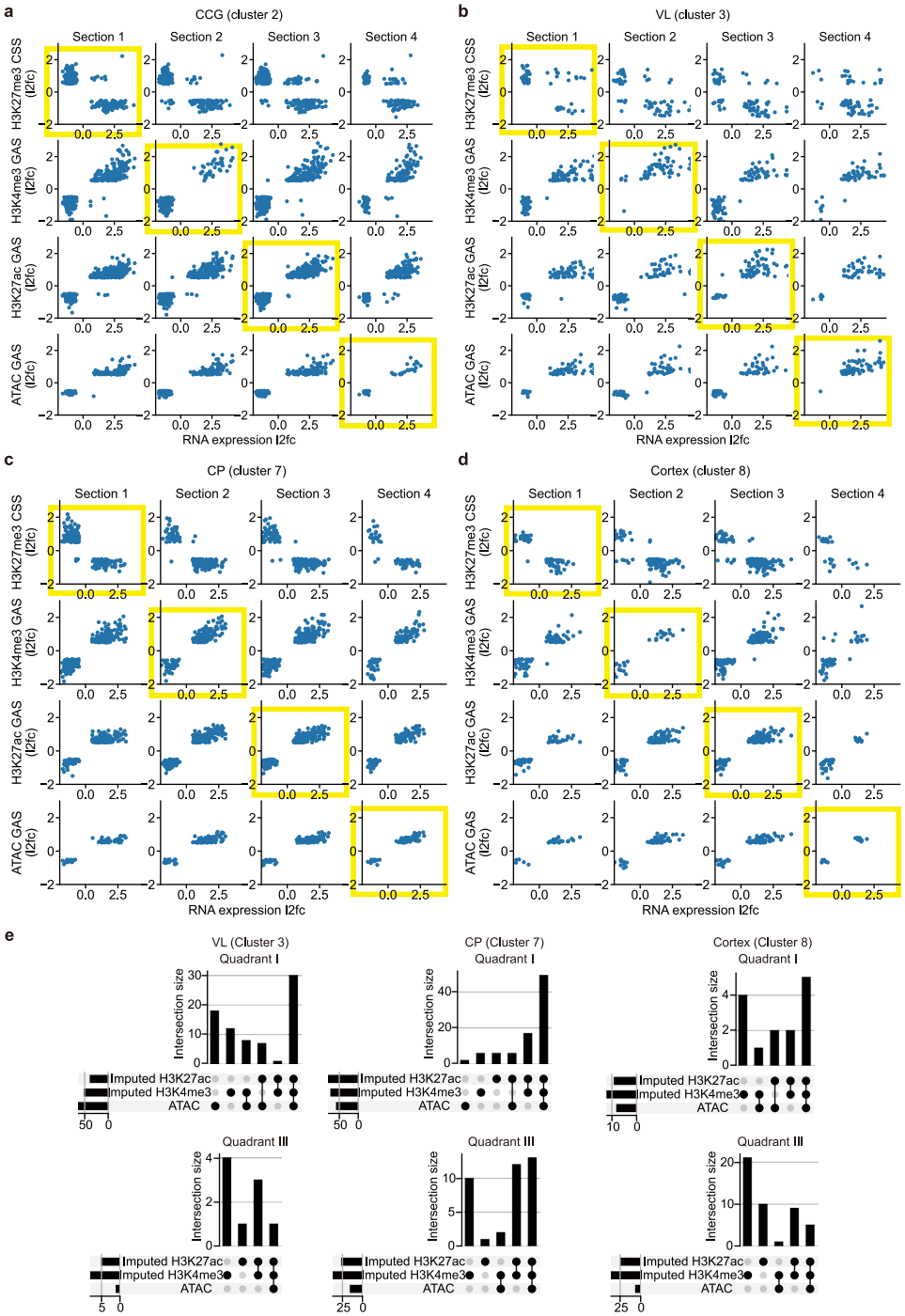
Extended Data Fig. 1 | Simulation settings and plots of results from baseline methods. a, Five sets of simulation parameters used for data generation. **b**, Mosaic configuration of each dataset. A dotted box denotes that the modality is missing. **c**, UMAP plots of RNA and protein profiles before batch correction. Spots are colored by section and factor labels for each modality. **d**, Spatial plots of clustering results from BANKSY and CellCharter applied to each modality. **e**, UMAP plots of embeddings from BANKSY and CellCharter applied to each modality. ADT, antibody-derived tags.

https://doi.org/10.1038/s41588-026-02573-3



Extended Data Fig. 2 | Quantitative evaluation of individual sections and plots of results from benchmarked methods. a, Spatial plots of the clustering results of benchmarked methods. **b**, UMAP plots of embeddings from benchmarked methods. From top to bottom, spots are colored by section, cluster, and factor labels. **c**, ARI scores of benchmarked methods across three section types. Box plots display the distribution of scores from five simulation cases, each with three replicates (n = 15 per box). Centerlines indicate the median; box limits represent the interquartile range (IQR); whiskers denote 1.5× IQR; dots indicate outliers beyond 1.5× IQR. **d**, PAS and CHAOS scores of benchmarked methods across three section types. Box plots display the distribution of scores from five simulation cases, each with three replicates (n = 15 per box). Centerlines indicate the median; box limits represent the interquartile range (IQR); whiskers denote 1.5 times the IQR (1.5× IQR); dots indicate outliers beyond 1.5× IQR.

https://doi.org/10.1038/s41588-026-02573-3



Extended Data Fig. 3 | Analyzing the correlation between RNA and GAS/CSS in selected brain regions. All genes plotted have an adjusted p value $<1\times10^{-5}$ and absolute value of $\log_2$ fold change >0.5. The yellow boxes in **a**–**d** denote the relationship between the measured modalities in each section. **a**, Correlation between RNA expression and GAS/CSS for CCG-specific (SpaMosaic's cluster 2) genes across the four sections. **b**, Correlation between RNA expression and GAS/CSS for VL-specific (SpaMosaic's cluster 3) genes across all sections. **c**, Correlation between RNA expression and GAS/CSS for CP-specific (SpaMosaic's cluster 7) genes across all sections. **d**, Correlation between RNA expression and GAS/CSS for the cortex-specific (SpaMosaic's cluster 8) genes across all sections. **e**, Upset plots of three region-specific gene sets in quadrants I (top) and III (bottom) based on the correlations between measured RNA and measured ATAC, imputed H3K4me3, and imputed H3K27ac for section 4.

https://doi.org/10.1038/s41588-026-02573-3



Extended Data Fig. 4 | Spatial plots of selected clusters and gene activity scores/chromatin silencing scores of region-specific marker genes. For each tissue panel, the top row displays the spatial distribution of selected clusters (colored by cluster identity). The bottom row shows spatial heatmaps representing the gene expression, gene activity scores or chromatin silencing scores of the corresponding region-specific marker genes.

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Last updated by author(s): Jan 27, 2026

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☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for data collection.
Data analysis	The SpaMosaic method (v1.0.3) is implemented in an open-source Python package that is available at: https://github.com/JinmiaoChenLab/SpaMosaic Additional software used in this manuscript: Python v3.8.8; Scanpy v1.9.6; numpy v1.22.4; pandas v1.5.3; h5py v3.10.0; seaborn v0.12.2; scikit-learn v1.3.2; scikit-misc v0.2.0; annoy (https://github.com/spotify/annoy); hnswlib (https://github.com/nmslib/hnswlib); scib v1.1.4; scanpy v1.9.6; squidpy v1.2.3; harmony-pytorch v0.1.7; ipykernel v6.27.0; jupyterlab v4.0.9; rpy2 v3.5.14; mclust v6.0.0; clusterProfiler v4.12.6; Signac v1.12.0; Space Ranger v2.1.0; MACS2 v2.2.6; torch v2.1.1+cu121; torch_geometric v2.4.0; pyg_lib v0.3.1+pt21cu121; torch_scatter v2.1.2+pt21cu121; torch_sparse v0.6.18+pt21cu121; torch_cluster v1.6.3+pt21cu121; torch_spline_conv v1.2.2+pt21cu121; CLUE (https://github.com/openproblems-bio/neurips2021_multimodal_topmethods/tree/main/src/match_modality/methods/clue); Cobolt v1.0.1; scmomat v0.2.0; StabMap v1.3.0 (https://github.com/MarioniLab/StabMap); MIDAS (https://github.com/labomics/midas); scvi-tools v0.19.0 (https://github.com/scverse/scvi-tools); babel (https://github.com/wukevin/babel); dance v1.0.1; MultiVI (https://github.com/scverse/scvi-tools); UINMF v1.0.1(https://github.com/welch-lab/liger/tree/v1.0.1); CellCharter v0.3.5(https://github.com/CSOgroup/cellcharter/tree/main); BANKSY v1.2.1(https://github.com/prabhakarlab/Banksy_py/tree/Bansky_1.2.1)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All datasets used in this study were obtained from public repositories, except for the in-house generated human lymph node and tonsil datasets. The Misar-seq sections were downloaded from https://www.biosino.org/node/project/detail/OEP003285 and https://pan.baidu.com/s/15i0R5na2Jud_7EpkWSaZpA?pwd=oipj. The Stereo-seq sections were obtained from https://db.cngb.org/stomics/mosta/download/. The spatial RNA-Epigenomics seq sections were accessed via https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205055 and https://cells.ucsc.edu/?ds=brain-spatial-omics. The Spatial ATAC-seq, Spatial Cut&Tag-seq, Spatial Mux-seq, and DBiT-ARP-seq sections were retrieved from NCBI GEO under the accession numbers GSE171943, GSE165217, GSE263333, and GSE308623, respectively. The mouse brain 10x Visium and Visium HD datasets were obtained from the publicly available 10x Genomics resources (https://www.10xgenomics.com/datasets/mouse-brain-section-coronal-1-standard and https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-libraries-of-mouse-brain-he). All processed datasets are publicly available at https://zenodo.org/records/16813383.

We used the scRNA-seq atlas of the mouse nervous system (available at http://mousebrain.org/adolescent/downloads.html) to assist in the analysis of cell types.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N.A.
Reporting on race, ethnicity, or other socially relevant groupings	N.A.
Population characteristics	N.A.
Recruitment	N.A.
Ethics oversight	N.A.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	For the public datasets, the original authors who made the datasets publicly available determined the sample size. For the in-house human lymph node dataset, three lymph node sections of 5 µm thickness were used from formalin-fixed, paraffin-embedded (FFPE) lymph node. For the in-house human tonsil dataset, three tonsil sections of 5 µm thickness were used from formalin-fixed, paraffin-embedded (FFPE) tonsil.
Data exclusions	In the integration experiments, we removed one modality measurement from sections with multiple modalities to simulate the mosaic dataset setting.
Replication	Not applicable. Our study did not aim to uncover any mechanistic or intervention effect. Instead, we benchmarked our proposed method against competing methods with different datasets acquired with various technologies.
Randomization	Not applicable. Our study did not aim to uncover any mechanistic or intervention effect and hence did not require any controls.
Blinding	Not applicable. Our study did not involve human participants and their responses.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	N.A.
Wild animals	N.A.
Reporting on sex	N.A.
Field-collected samples	N.A.
Ethics oversight	N.A.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	N.A.
Novel plant genotypes	N.A.
Authentication	N.A.